

Carry periodicity forces a large $(x^3 + 1)$ factor: a proof of Karttunen's conjecture on the binary columns of the Fibonacci numbers

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Abstract

Write the Fibonacci numbers one under another in binary. The n -th column from the right is periodic in k with period $3 \cdot 2^n$; reading one full period as the coefficient vector of a polynomial over $\text{GF}(2)$ gives OEIS A036284. In November 1998 A. Karttunen conjectured that this polynomial is divisible by $(x^3 + 1)^{2^{n-1}-1}$. We prove it. The whole content is that the *carry* into position n depends only on the low n bits of the two summands and therefore has period $3 \cdot 2^{n-1}$ — exactly half the period of the column itself. Since $x^{3 \cdot 2^m} + 1 = (x^3 + 1)^{2^m}$ in characteristic 2, halving the period is the same as extracting a factor $(x^3 + 1)^{2^{n-1}}$, and the conjecture drops out of a one-line valuation count. The argument in fact gives the exact $(x^2 + x + 1)$ -adic valuation, not just the lower bound.

1 The sequence and the conjecture

Let $F(0) = 0$, $F(1) = 1$, $F(k) = F(k-1) + F(k-2)$. Writing the $F(k)$ in binary, one under another, the bits in the n -th column from the right repeat with period $3 \cdot 2^n$. OEIS A036284, contributed by Antti Karttunen on 1 November 1998 under the name *Periodic vertical binary vectors of Fibonacci numbers*, is one full period of that column read as a binary number:

$$a(n) = \sum_{k=0}^{3 \cdot 2^n - 1} \left(\lfloor F(k)/2^n \rfloor \bmod 2 \right) 2^k, \quad a(0), a(1), a(2), a(3) = 6, 24, 1440, 5728448.$$

Identifying an integer with the $\text{GF}(2)$ polynomial whose coefficient of x^k is bit k , the entry carries the comment

Conjecture: For $n \geq 1$, each term $a(n)$, when considered as a $\text{GF}(2)[X]$ -polynomial, is divisible by the $\text{GF}(2)[X]$ -polynomial $(x^3 + 1)^{A000225(n-1)}$

Conjecture 2: there is also one extra $(x^1 + 1)$ factor present.

Here $A000225(n-1) = 2^{n-1} - 1$. This note proves the first conjecture. Conjecture 2 is proved in a companion note by the same lemma, read on the other irreducible factor. As of the “Last modified” line on the live entry (23 August 2026) both statements are still recorded as unproven conjectures, no proof appears in the entry's links or programs, and a literature search on binary columns of the Fibonacci numbers turns up nothing.

2 Notation

Throughout $n \geq 1$ is fixed and

$$L = 3 \cdot 2^n, \quad q(x) = x^2 + x + 1, \quad x^3 + 1 = (x + 1)q(x).$$

For a nonzero $P \in \text{GF}(2)[x]$ and an irreducible f , write $v_f(P)$ for the exponent of f in P . Two facts are used repeatedly:

$$x^{3 \cdot 2^m} + 1 = (x^3 + 1)^{2^m} \quad \text{in } \text{GF}(2)[x] \quad (m \geq 0), \quad (1)$$

by the Frobenius endomorphism, and $v_q((x^3 + 1)^m) = v_{x+1}((x^3 + 1)^m) = m$.

Let $r_k = F(k) \bmod 2^{n+1}$, let

$$b(k) = \lfloor r_k / 2^n \rfloor \in \{0, 1\}$$

be the n -th bit, and let $s_k = r_k \bmod 2^n$ be the low part, so that $r_k = b(k)2^n + s_k$. Finally set

$$\gamma(k) = \lfloor (s_{k-1} + s_{k-2}) / 2^n \rfloor \in \{0, 1\},$$

the carry into position n when $F(k-1)$ and $F(k-2)$ are added. Put

$$A(x) = \sum_{k=0}^{L-1} b(k) x^k,$$

so that A is the polynomial attached to $a(n)$.

3 Three lemmas

Lemma 1. *For every $m \geq 1$ the sequence $(F(k) \bmod 2^m)_{k \geq 0}$ is purely periodic with least period $3 \cdot 2^{m-1}$.*

Proof. Pure periodicity holds for any modulus because the recurrence is invertible: $F(k-2) = F(k) - F(k-1)$. The value $\pi(2^m) = 3 \cdot 2^{m-1}$ of the Pisano period at prime power 2^m is classical; see Wall [1]. $\square$

In particular $(r_k)_k$ has period $L = 3 \cdot 2^n$, so b , s , γ are all well defined as functions on $\mathbb{Z}/L\mathbb{Z}$, and $r_k \equiv r_{k-1} + r_{k-2} \pmod{2^{n+1}}$ holds for *every* $k \in \mathbb{Z}/L\mathbb{Z}$ (for $k \geq 2$ it is the recurrence; for $k = 0, 1$ it follows by periodicity from the instances $k = L, L+1$).

Lemma 2 (carry identity). *For every $k \in \mathbb{Z}/L\mathbb{Z}$,*

$$b(k) + b(k-1) + b(k-2) \equiv \gamma(k) \pmod{2}.$$

Proof. Split both summands into high bit and low part:

$$r_{k-1} + r_{k-2} = (b(k-1) + b(k-2))2^n + (s_{k-1} + s_{k-2}) = (b(k-1) + b(k-2) + \gamma(k))2^n + (s_{k-1} + s_{k-2}) \bmod 2^n,$$

using $s_{k-1} + s_{k-2} = \gamma(k)2^n + ((s_{k-1} + s_{k-2}) \bmod 2^n)$, which is legitimate because $s_{k-1} + s_{k-2} < 2^{n+1}$. Reducing modulo 2^{n+1} and comparing with $r_k = b(k)2^n + s_k$, where $s_k = (s_{k-1} + s_{k-2}) \bmod 2^n < 2^n$, forces $b(k) \equiv b(k-1) + b(k-2) + \gamma(k) \pmod{2}$. $\square$

Lemma 3 (the carry has half the period). $\gamma(k + 3 \cdot 2^{n-1}) = \gamma(k)$ for every k .

Proof. By definition $\gamma(k)$ is a function of s_{k-1} and s_{k-2} alone, and $s_j = F(j) \bmod 2^n$. By Lemma 1 with $m = n$ the sequence $(s_j)_j$ has period $3 \cdot 2^{n-1}$. $\square$

This is the whole point: the column b has period $L = 3 \cdot 2^n$, while the carry γ that drives it has period only $L/2$.

4 The theorem

Theorem 4. *Let $n \geq 1$ and let $\widehat{C}(x) = \sum_{k=0}^{L/2-1} \gamma(k)x^k$. Then in $\text{GF}(2)[x]$*

$$q(x)A(x) = \widehat{C}(x)(x^3 + 1)^{2^{n-1}} + M(x)(x^3 + 1)^{2^n} \quad (2)$$

for some $M \in \text{GF}(2)[x]$. Consequently

$$v_q(A) \geq 2^{n-1} - 1, \quad v_{x+1}(A) \geq 2^{n-1},$$

and therefore $(x^3 + 1)^{2^{n-1}-1}$ divides A .

Proof. Multiply the identity of Lemma 2 by x^k and sum over $k \in \mathbb{Z}/L\mathbb{Z}$. On the left, reading indices modulo L turns the shift $k \mapsto k - 1$ into multiplication by x modulo $x^L + 1$, so the left side becomes $(1 + x + x^2)A(x) = q(x)A(x)$ modulo $x^L + 1$. On the right we get $\Gamma(x) = \sum_{k=0}^{L-1} \gamma(k)x^k$. Hence

$$q(x)A(x) \equiv \Gamma(x) \pmod{x^L + 1}.$$

By Lemma 3 the coefficient list of Γ is the concatenation of two copies of that of $\widehat{C}$, i.e.

$$\Gamma(x) = \widehat{C}(x)(1 + x^{L/2}) = \widehat{C}(x)(x^{3 \cdot 2^{n-1}} + 1) = \widehat{C}(x)(x^3 + 1)^{2^{n-1}},$$

the last step by (1). Since $x^L + 1 = (x^3 + 1)^{2^n}$, again by (1), this is exactly (2).

Now take valuations in (2). Both terms on the right are divisible by $(x^3 + 1)^{2^{n-1}}$, hence

$$v_q(qA) \geq 2^{n-1}, \quad v_{x+1}(qA) \geq 2^{n-1}.$$

Since $v_q(q) = 1$ and $v_{x+1}(q) = 0$, this reads $v_q(A) \geq 2^{n-1} - 1$ and $v_{x+1}(A) \geq 2^{n-1}$. Finally

$$v_{x^3+1}(A) = \min(v_{x+1}(A), v_q(A)) \geq 2^{n-1} - 1. \quad \square$$

Corollary 5. *For every $n \geq 1$, $(x^3 + 1)^{2^{n-1}-1}$ divides $a(n)$ in $\text{GF}(2)[X]$. This is Karttunen's conjecture.*

Remark (the exact valuation). Equation (2) gives more than the bound. If $v_q(\widehat{C}) < 2^{n-1}$ then the two right-hand terms have different q -valuations and

$$v_q(A) = 2^{n-1} - 1 + v_q(\widehat{C}),$$

and likewise $v_{x+1}(A) = 2^{n-1} + v_{x+1}(\widehat{C})$ when $v_{x+1}(\widehat{C}) < 2^{n-1}$. Computation (Section 5) gives $v_q(\widehat{C}) = 0$ for $1 \leq n \leq 12$, so $v_q(A) = 2^{n-1} - 1$ exactly there — the conjectured bound is sharp on the q -side. On the $(x+1)$ -side $v_{x+1}(\widehat{C}) = 0$ for $n = 1$ and $4 \leq n \leq 12$, but $v_{x+1}(\widehat{C}) = 1$ at $n = 2$ and 3 at $n = 3$, which is why $v_{x+1}(A)$ equals 3 and 7 there instead of 2 and 4. I have *not* proved that $v_q(\widehat{C}) = 0$ for all n ; that statement is equivalent to $\widehat{C}(\omega) \neq 0$ for ω a primitive cube root of unity in $\text{GF}(4)$, i.e. to a condition on the distribution modulo 3 of the positions of the carries. It is an open question here.

Remark (what the proof really uses). Nothing about the Fibonacci recurrence is used except that the modulus- 2^m period doubles when m increases by one. The same argument gives, verbatim, the analogous statement for any integer sequence satisfying $u(k) = u(k-1) + u(k-2)$ whose period modulo 2^m is $\pi \cdot 2^{m-1}$ for a fixed odd π : the column polynomial is divisible by $(x^\pi + 1)^{2^{m-1}-1}$. The proof is elementary and the natural destination is an OEIS comment, not a journal.

5 Verification

A verification script, written separately from this note, checks every assertion made here and prints every number quoted. It (i) recomputes $a(0), \dots, a(3)$ from the defining sum and matches the published values 6, 24, 1440, 5728448 exactly; (ii) checks Lemma 2 at all L residues for $n = 0, \dots, 16$, that is up to $L = 196608$, with no failure; (iii) checks Lemma 3 over the same range, with no failure; (iv) verifies the identity (2) by explicit polynomial division for $n = 1, \dots, 12$; (v) computes the valuations, obtaining $v_q(A) = 2^{n-1} - 1$ and $v_{x+1}(A) = 2^{n-1}$ for $n = 1$ and $4 \leq n \leq 12$, and the two exceptional values $v_{x+1}(A) = 3$ at $n = 2$ and 7 at $n = 3$; and (vi) divides $a(n)$ by $(x^3 + 1)^{2^{n-1}-1}$ directly for $n = 1, \dots, 12$, obtaining remainder zero in every case.

References

- [1] D. D. Wall, *Fibonacci series modulo m*, Amer. Math. Monthly **67** (1960), 525–532.
- [2] R. Lidl and H. Niederreiter, *Finite Fields*, 2nd ed., Cambridge University Press, 1997.
- [3] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A036284>, entry A036284, consulted 23 August 2026.

A Suggested OEIS comment (plain text, house style)

Both conjectures are true, and the reason is carry periodicity. Fix $n \geq 1$, put $L = 3 \cdot 2^n$, let $b(k)$ be bit n of $F(k)$ and let $g(k)$ be the carry into bit n when $F(k-1)$ and $F(k-2)$ are added, i.e. $g(k) = \text{floor}(((F(k-1) \bmod 2^n) + (F(k-2) \bmod 2^n)) / 2^n)$. Splitting each summand into its bit n and its low n bits gives $b(k) = b(k-1) + b(k-2) + g(k) \bmod 2$ for every k in $\mathbb{Z}/L$, since $F(k) = F(k-1) + F(k-2) \bmod 2^{n+1}$ and the Fibonacci sequence mod 2^{n+1} is purely periodic with period L . Multiplying by x^k and summing over $\mathbb{Z}/L$, $(x^2+x+1) A(x) = G(x) \bmod x^L + 1$, where A is the column polynomial (this entry) and G is the carry polynomial. But $g(k)$ depends only on $F(k-1)$ and $F(k-2) \bmod 2^n$, whose period is $L/2$, so $G(x) = C(x) \cdot (1 + x^{L/2})$ with C the first half of G . In characteristic 2, $1 + x^{L/2} = (x^3+1)^{2^{n-1}}$ and $x^L + 1 = (x^3+1)^{2^n}$. Hence

$$(x^2+x+1) A(x) = C(x) (x^3+1)^{2^{n-1}} + M(x) (x^3+1)^{2^n}.$$

Both right-hand terms are divisible by $(x^3+1)^{2^{n-1}}$. Cancelling the single factor x^2+x+1 on the left gives (x^2+x+1) -multiplicity at least $2^{n-1} - 1$, while $x+1$ does not divide x^2+x+1 , so the $(x+1)$ -multiplicity is at least 2^{n-1} . Together: $(x^3+1)^{2^{n-1}-1}$ divides $a(n)$, with one extra $(x+1)$ factor. The bound is exact whenever C is not divisible by the corresponding factor; the $(x+1)$ count is 2^{n-1} for $n = 1$ and $n \geq 4$, and $2^{n-1} + 1$ and $2^{n-1} + 3$ at $n = 2$ and $n = 3$.